Algorithms and error correction

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Outline

Part	Main question
Quantum rules and circuits	How do qubits behave?
Quantum algorithms	Where can interference help?
Quantum error correction	How can we protect a state?
Toric code and experiments	How can local checks protect global information?
Outlook	What remains unclear?

Classical bits and qubits

Probabilistic classical bit

Values

$0,1$

Probabilities

$P(0) = p_1, P(1) = p_2$

$p_1 + p_2 = 1, p_1, p_2 \geq 0$

Gate: bit flip

$0 \leftrightarrow 1, p_1 \leftrightarrow p_2$

Quantum bit (qubit)

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{C}^2$

Probabilities

$P(0) = |\alpha|^2, P(1) = |\beta|^2$

$|\alpha|^2 + |\beta|^2 = 1$

Relative phase

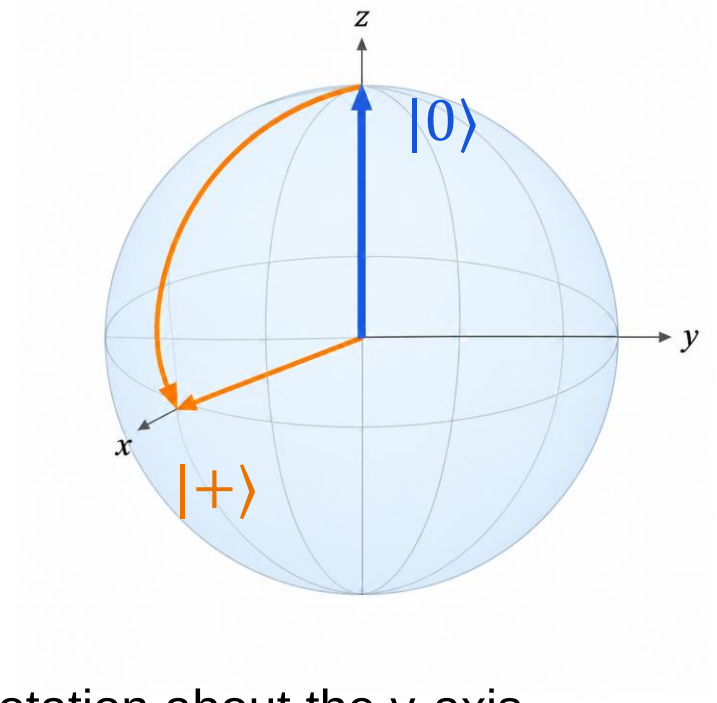
$\frac{\beta}{\alpha} = \left|\frac{\beta}{\alpha}\right| e^{i\theta}, \alpha \neq 0$

Quantum information includes relative phase.

Single-qubit gates as rotations

Unitary gates rotate a qubit's Bloch vector.

Example: a 90° rotation



Rotation about the y-axis

$$R_y(\frac{\pi}{2})|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Single-qubit universality

Exact with adjustable angles

$$U = e^{i\delta} R_z(\alpha) R_y(\beta) R_z(\gamma)$$

Rotations about two axes can realize any single-qubit gate.

Finite set: H and T

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Relative phase has observable consequences

Two states can have the same computational-basis probabilities

State	Measure directly	Apply H, then measure
$\|+\rangle = \frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	0 or 1, each 50%	0 with certainty
$\|-\rangle = \frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$	0 or 1, each 50%	1 with certainty

Postulate 1: quantum states

A pure state is described by a normalized state vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

The basis states are possible distinguishable outcomes.

Amplitudes describe both magnitude and relative phase.

An overall phase leaves physical predictions unchanged.

Postulate 2: reversible evolution

An isolated system evolves base on the Schrödinger equation

$$\mathrm{i}\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$$

so that $$|\psi(t)\rangle = \mathrm{e}^{-iHt/\hbar}|\psi(t=0)\rangle = U|\psi(t=0)\rangle.$$

A quantum gate preserves the length of the state vector.

It acts linearly on every superposition.

Running its inverse reverses the operation.

Postulate 3: measurement

Computational-basis measurement follows the Born rule

$$P(0) = |\alpha|^2 \qquad P(1) = |\beta|^2$$

Outcome b leaves the state $|b\rangle$

One run gives one classical outcome.

Many repeated preparations estimate the probabilities.

A measurement basis specifies which question we ask.

Example: $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ gives 0 with probability $\frac{3}{4}$.

Postulate 4: composite systems

Joint systems use a tensor-product state space

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Two qubits have four basis states.

n qubits have 2^n basis amplitudes.

One measurement still yields only one n-bit string.

Entanglement

Some joint states cannot separate into individual qubit states

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Each individual readout looks random.

Joint readouts are perfectly correlated in this basis.

No product $|a\rangle \otimes |b\rangle$ equals this joint pure state.

Entanglement will let us spread one logical state across many qubits.

The four postulates in one view

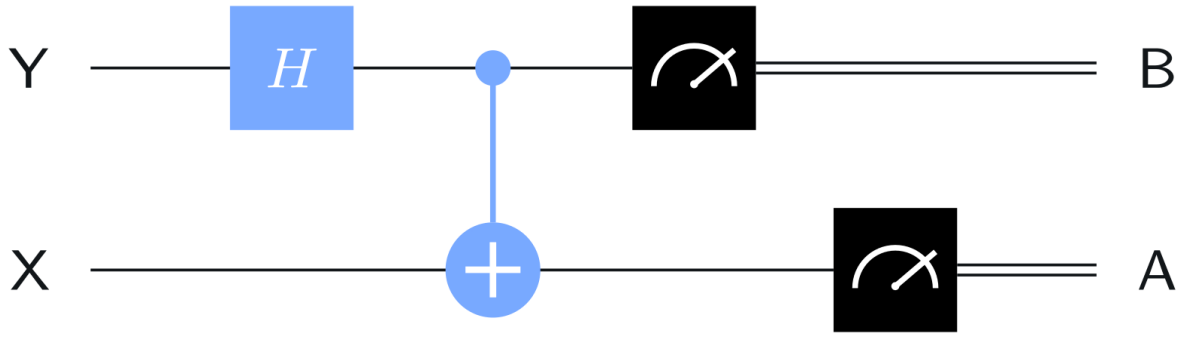
These rules define the circuit model used in this lecture

Ingredient	Operational meaning
State	A normalized vector describes a pure state
Evolution	Unitary gates transform amplitudes linearly
Measurement	Born probabilities and conditional state update
Composition	Tensor products describe joint systems

Open systems and noisy operations extend this ideal description.

Reading a quantum circuit

A circuit is an ordered sequence of gates and measurements



Each horizontal wire represents a qubit.

Read time from left to right.

A box acts on its wire. A connected symbol couples qubits.

A final measurement converts quantum information into classical bits.

Three useful gates

A gate is fully specified by its action on a basis

Gate	Action on $\|0\rangle$	Action on $\|1\rangle$
X (bit flip)	$\|1\rangle$	$\|0\rangle$
Z (phase flip)	$\|0\rangle$	$-\|1\rangle$
H (Hadamard)	$\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

$H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$. In particular, $H^2 = I$.

A one-qubit interference experiment

A phase change between two Hadamard gates changes the output

$$HH|0\rangle = |0\rangle$$

$$HZH|0\rangle = |1\rangle$$

First H creates equal amplitudes.

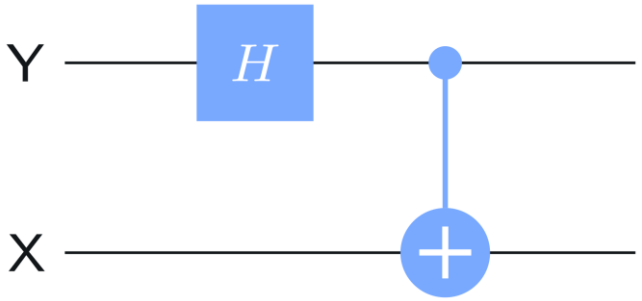
Z changes their relative sign.

Final H combines them into a definite outcome.

Interference converts phase information into a measurable answer.

A circuit that creates entanglement

Hadamard followed by controlled-NOT



Prepare two qubits in $|00\rangle$.

H on the control gives $\frac{|00\rangle+|10\rangle}{\sqrt{2}}$.

CNOT flips the target only when the control is 1.

Output: $\frac{|00\rangle+|11\rangle}{\sqrt{2}}$

What a quantum algorithm must accomplish

Available operations	Algorithm design
Prepare a simple state.	Encode a useful property in phases.
Apply an efficient sequence of gates.	Arrange constructive and destructive interference.
Sample the final measurement.	Extract the property with high probability.

A speedup depends on the problem, access model and classical comparison.

The Deutsch–Jozsa problem

Promise: f maps n-bit strings to one bit and is constant or balanced

Input x	Constant example	Balanced example
00	0	0
01	0	1
10	0	1
11	0	0

Question: which kind of function do we have? Count calls to the oracle.

The exact classical query cost

A deterministic classical algorithm may keep seeing the same answer

Worst case: $2^{n-1} + 1$ queries

After seeing half the inputs, equal outputs remain ambiguous.

The function could still be constant.

Or every unseen output could be the opposite value.

For $n = 2$, two matching answers are inconclusive. A third query decides.

A reversible oracle and phase kickback

Quantum access must preserve reversibility

$$U|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

$$U|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$$

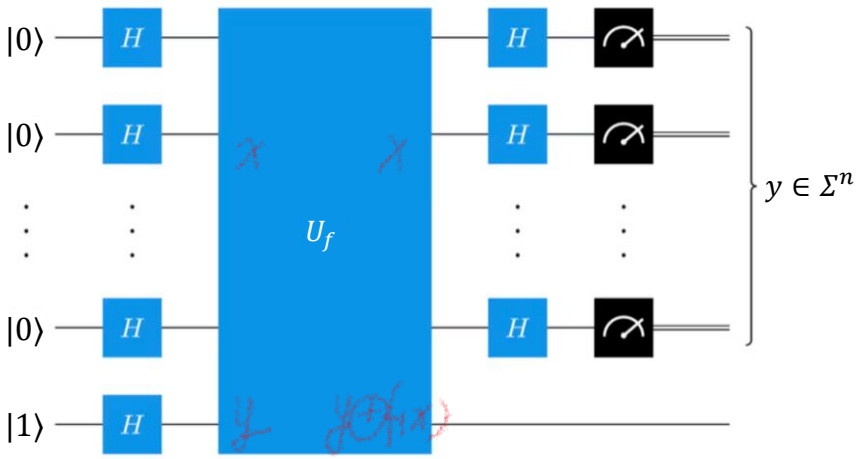
$\oplus$ means addition modulo 2.

The target $|-\rangle$ turns the answer bit into a sign.

That sign can interfere across different inputs x.

The Deutsch–Jozsa circuit

One coherent oracle call



H gates prepare an equal superposition of all x.

The oracle writes f(x) into relative signs.

Final H gates recombine the input amplitudes.

Measure $0 \ldots 0 \Leftrightarrow f$ is constant

The conclusion assumes the constant-or-balanced promise and ideal gates.

A two-bit example

Balanced function: $f(x_1x_2) = x_1 \oplus x_2$

x	00	01	10	11
$f(x)$	0	1	1	0
Phase $(-1)^{f(x)}$	$+1$	-1	-1	$+1$

Before final H gates: $|-\rangle \otimes |-\rangle$

The final state is $|11\rangle$, so the answer is balanced.

The scope of the Deutsch–Jozsa advantage

The query comparison depends on the required certainty

Method	Oracle queries	Guarantee
Quantum, ideal	1	Exact under the promise
Classical, deterministic	$2^{n-1}+1$	Exact in the worst case
Classical, randomized	k samples	Error at most 2^{1-k}

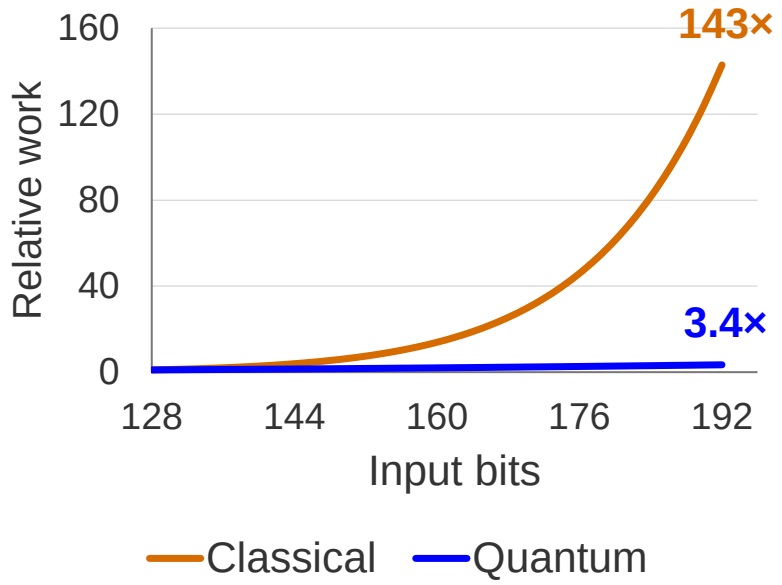
This is a lesson in interference and exact query complexity.

Quantum speedups: Shor and Grover

Shor: integer factoring

Factor an n-bit integer

Each model normalized at 128 bits



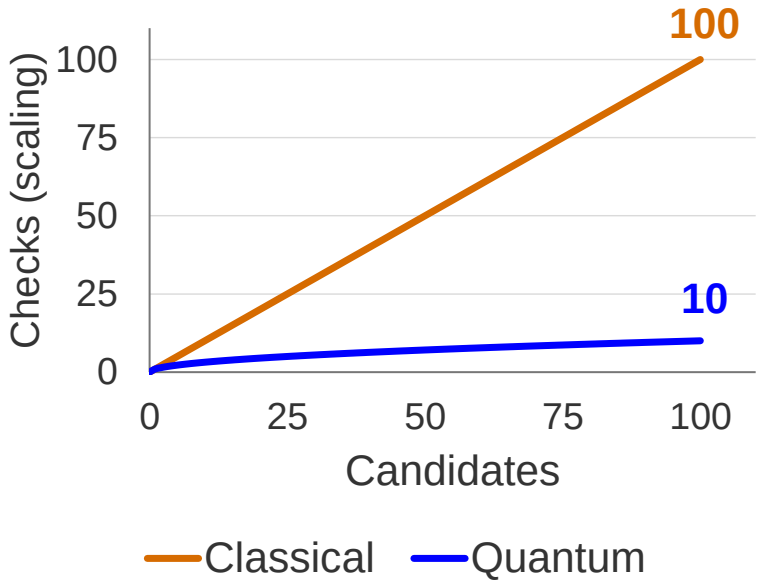
$T_Q = O(n^3)$

Superpolynomial advantage over best known classical methods

Grover: unstructured search

Find one marked item among N

One marked item, constants omitted



$O(N)$ vs $O(\sqrt{N})$

A million candidates: about a thousand quantum checks

Why quantum error correction is necessary

A long computation creates many opportunities for faults

$$P(\text{no fault}) = (1-p)^G$$

p: physical error rate

G: number of fault opportunities

Gates, idle periods and readout can all introduce noise.

For $p = 10^{-3}$ and $G = 10^5$,$(1-p)^G \approx 3.5 \times 10^{-44}$.

Without error correction, a quantum computer may be useless for practical applications.

The formula is a toy model of independent faults, not an algorithm success rate.

Bit flips and phase flips

Pauli errors give a useful language for qubit noise

Error	Effect on $\alpha\|0\rangle + \beta\|1\rangle$	What changes?
X	$\alpha\|1\rangle + \beta\|0\rangle$	Bit values swap
Z	$\alpha\|0\rangle - \beta\|1\rangle$	Relative phase flips
$Y = iXZ$	Both, up to an overall phase	Bit and phase

General single-qubit errors can be expanded in I, X, Y and Z.

Classical redundancy

The repetition code stores one bit in three physical bits

Logical bit	Stored word	One possible corrupted word
0	000	010
1	111	101

Majority vote corrects one bit flip. Can we do this for a qubit?

The quantum constraints

Unknown states cannot be cloned

A universal device cannot produce $|\psi\rangle \otimes |\psi\rangle$ from one unknown $|\psi\rangle$.

Linearity prevents the classical copying rule from working for every superposition.

Direct readout reveals too much

Measuring data qubits can reveal the encoded value.

That information destroys an unknown superposition.

The solution is an entangled encoding and measurements of relationships.

Encoding one logical qubit

The three-qubit bit-flip code

$$|0_l\rangle = |000\rangle \qquad |1_l\rangle = |111\rangle$$

$$|\psi_l\rangle = \alpha|000\rangle + \beta|111\rangle$$

The logical amplitudes α and β remain unknown. Parity checks are Z_1Z_2 and Z_2Z_3.

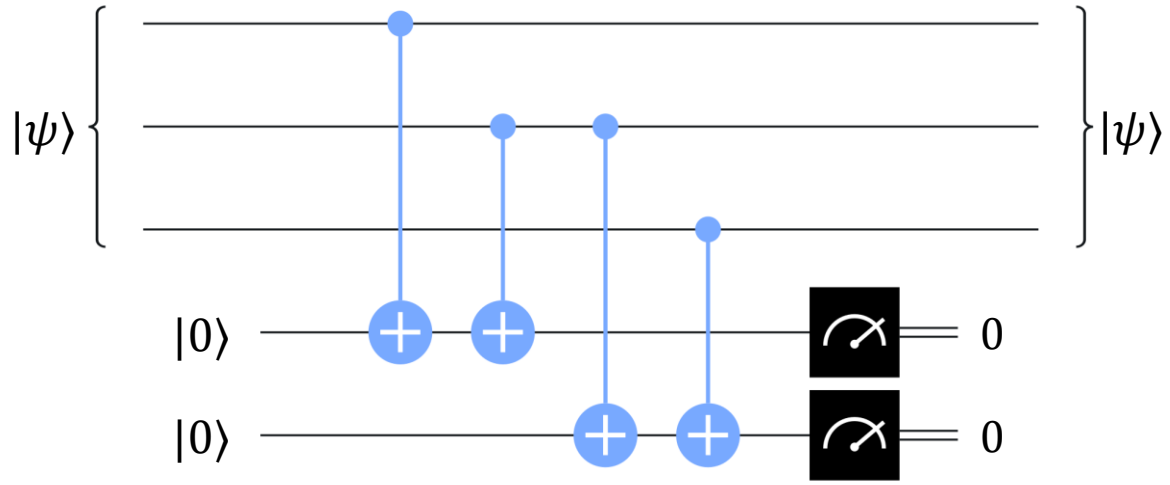
The information lives in a joint state.

This code protects against one X error.

The encoded state is generally entangled. It is not three independent copies.

A parity check reads a relationship

A fresh ancilla can measure $Z_1 Z_2$



$$|a_1 a_2 a_3\rangle|00\rangle \rightarrow |a_1 a_2 a_3\rangle|a_1 \oplus a_2 \; a_2 \oplus a_3\rangle$$

Equal bits give parity 0, or check value $+1$.

Different bits give parity 1, or check value -1.

Both $|000\rangle$ and $|111\rangle$ have the same valid parity.

The ancilla learns error information without reading α or β.

The syndrome locates a bit flip

Measure the commuting checks Z_1Z_2 and Z_2Z_3

Fault	Z_1Z_2	Z_2Z_3	Correction
None	$+1$	$+1$	I
X_1	-1	$+1$	X_1
X_2	-1	-1	X_2
X_3	$+1$	-1	X_3

Example: $\alpha|010\rangle + \beta|101\rangle$ gives $(-1,-1)$. Apply X_2.

Stabilizers protect both kinds of information

A stabilizer is a check that every valid code state passes

$$S|\psi_i\rangle = |\psi_i\rangle$$

Z-type checks reveal X errors.

X-type checks reveal Z errors.

All checks must mutually commute.

A -1 check outcome signals a defect. The full pattern is the syndrome.

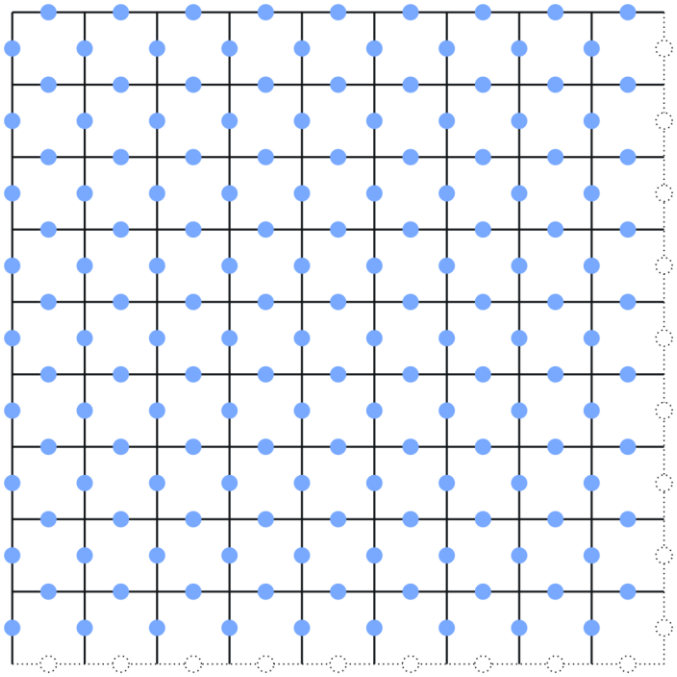
The error-correction cycle

Quantum data and classical decoding work together

Step	Role
1. Measure checks	Ancillas extract error syndromes
2. Decode	A classical algorithm estimates a likely error
3. Recover or track	Apply a correction or update a Pauli frame
4. Repeat	Reset ancillas and continue protecting the data

Toric code: a periodic square lattice

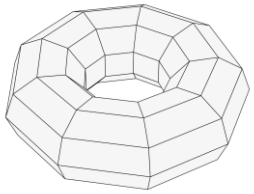
One data qubit sits on each edge



Identify the left and right edges.

Identify the top and bottom edges.

An $L \times L$ lattice contains $2L^2$ data qubits.

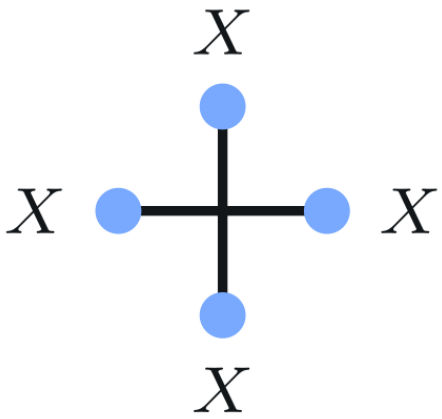


The torus specifies connectivity. The processor need not have a doughnut shape.

Illustration: IBM Quantum Learning

Star checks

Each vertex defines an X-type check



X stabilizer generator (vertex)

$$A_v = X_1 X_2 X_3 X_4$$

Four edges meet at the vertex v.

The check returns $+1$ or -1.

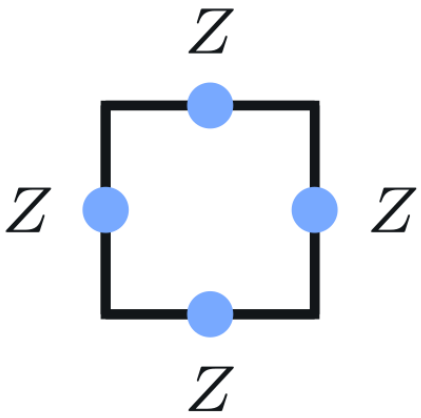
Code states have $A_v = +1$.

A star tests a collective X-parity of four nearby data qubits.

Illustration: IBM Quantum Learning

Plaquette checks

Each square face defines a Z-type check



Z stabilizer generator (tile)

$$B_p = Z_1 Z_2 Z_3 Z_4$$

Four edge qubits surround the face p.

The check returns $+1$ or -1.

Code states have $B_p = +1$.

Star checks detect Z errors. Plaquette checks detect X errors.

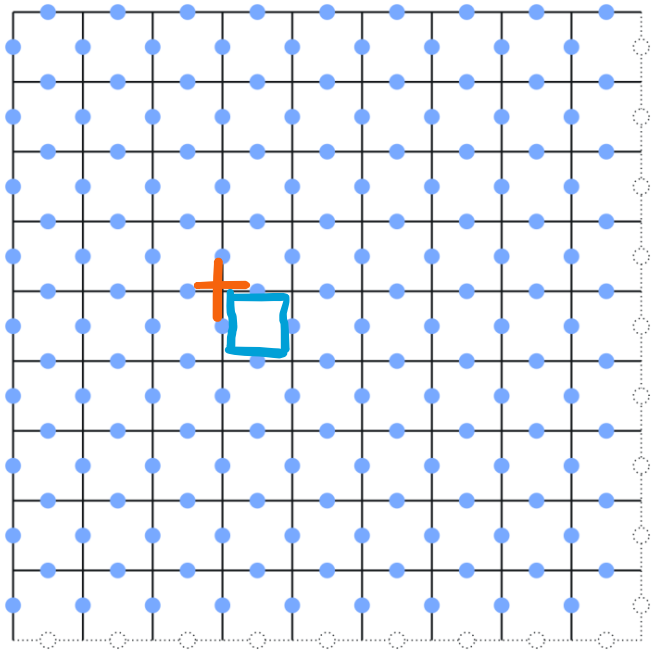
Illustration: IBM Quantum Learning

Why both check families are compatible

An adjacent star and plaquette share two data qubits

$$XZ = -ZX$$

$$A_v B_p = (-1)^2 B_p A_v = B_p A_v$$



Each shared qubit contributes one minus sign.

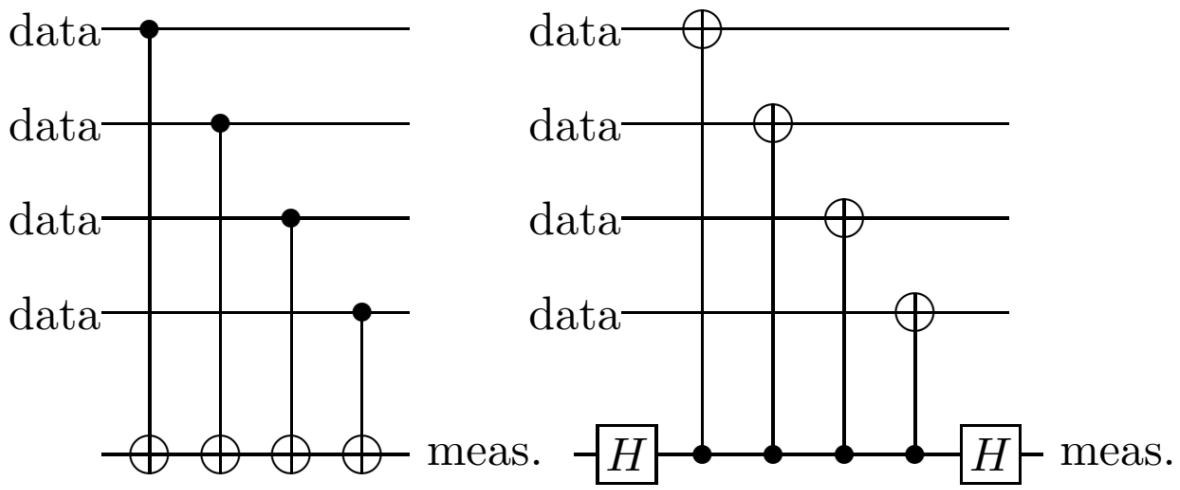
The two signs cancel.

All checks can have definite values at the same time.

The code space is the simultaneous +1 eigenspace of every check.

Measuring a four-qubit check

An ancilla reads one collective parity



$ZZZZ$: data qubits control CNOTs into a $|0\rangle$ ancilla.

XXXX: a $|+\rangle$ ancilla controls CNOTs on the data.

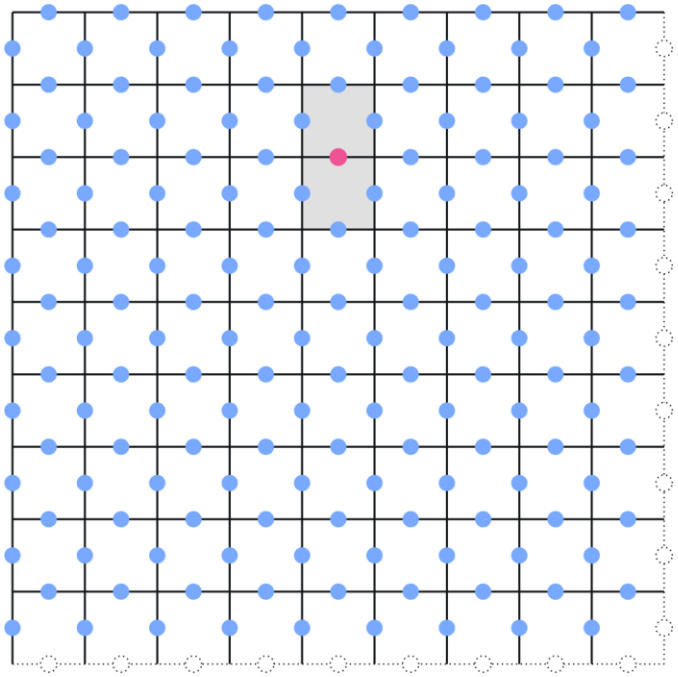
Measure the ancilla in Z or X, respectively.

This is a conceptual check circuit. Fault-tolerant scheduling also matters.

Dennis et al. (2002), Fig. 14. Ancillas start in $|0\rangle$

One bit error creates two defects

An X error acts on one edge qubit



The two neighboring plaquettes change sign to -1.

All other plaquette checks remain $+1$.

Star checks remain $+1$.

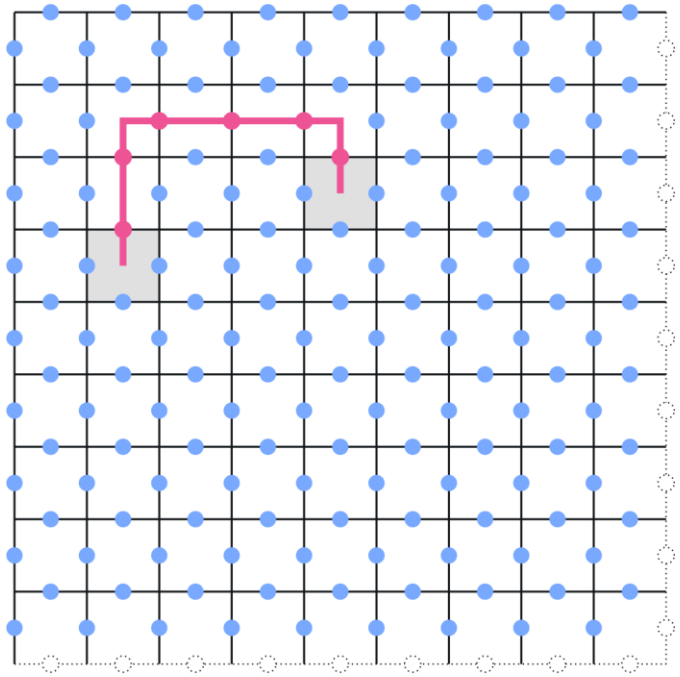
Pink: X errors Gray: −1 plaquettes

The syndrome is the pattern of failed checks, not a measurement of the data.

Illustration: IBM Quantum Learning

An error chain leaves only endpoints

Extend the X error along a path on the dual lattice



An interior plaquette touches two X errors.

Its check sign flips twice and returns to $+1$.

Only the two endpoint plaquettes remain defective.

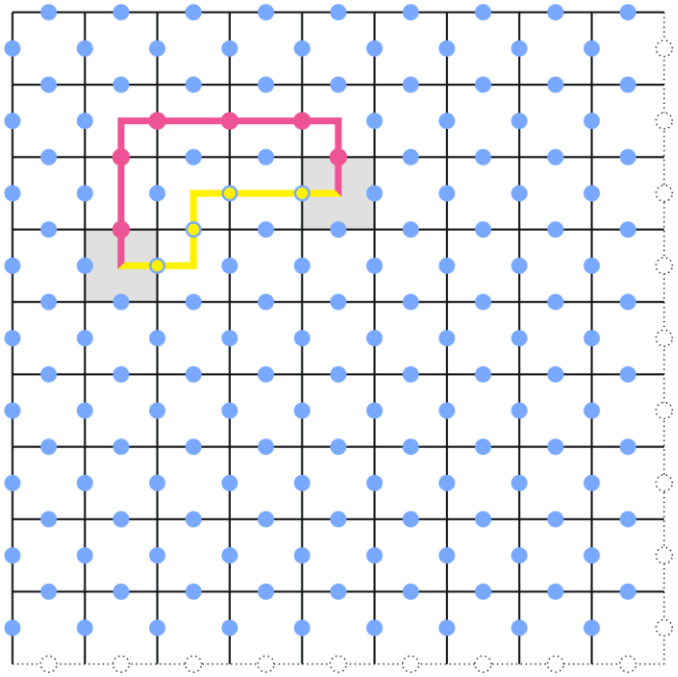
Pink: X errors Gray: −1 plaquettes

The syndrome reveals endpoints. Many different paths give the same syndrome.

Illustration: IBM Quantum Learning

Decoding is a classical graph problem

A decoder pairs defects with plausible recovery paths



Pink marks the unknown X-error path.

Yellow marks a candidate correction.

Their combination must preserve the logical state.

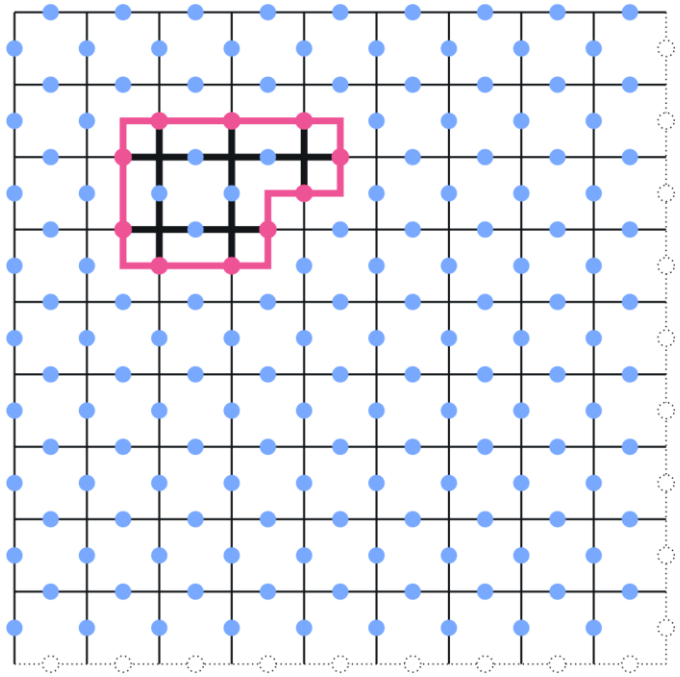
Yellow: proposed X correction

For sparse noise, short paths are plausible. Matching is one decoding method.

Illustration: IBM Quantum Learning

Contractible loops are stabilizers

A dual loop that can shrink away preserves every encoded state



Multiply the star checks enclosed by the dual loop.

Interior X factors appear twice and cancel: $X^2 = I$.

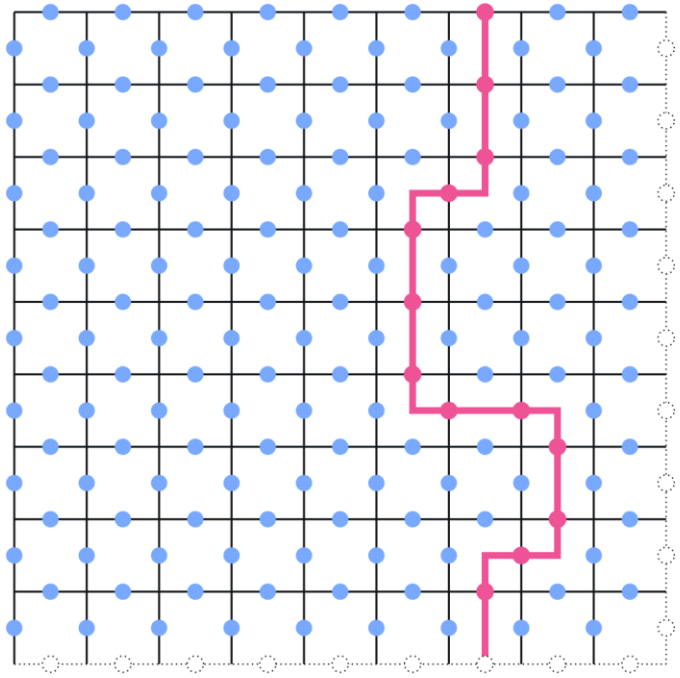
Only the boundary X loop remains.

Local deformation changes a dual path while preserving its logical action.

Illustration: IBM Quantum Learning

Winding loops and logical errors

A dual loop around the torus also has no endpoints



Every local check can still return $+1$.

The loop cannot shrink away using star checks.

Its action is a logical X operation.

No syndrome does not guarantee that the logical state is unchanged.

Illustration: IBM Quantum Learning

A worked logical failure

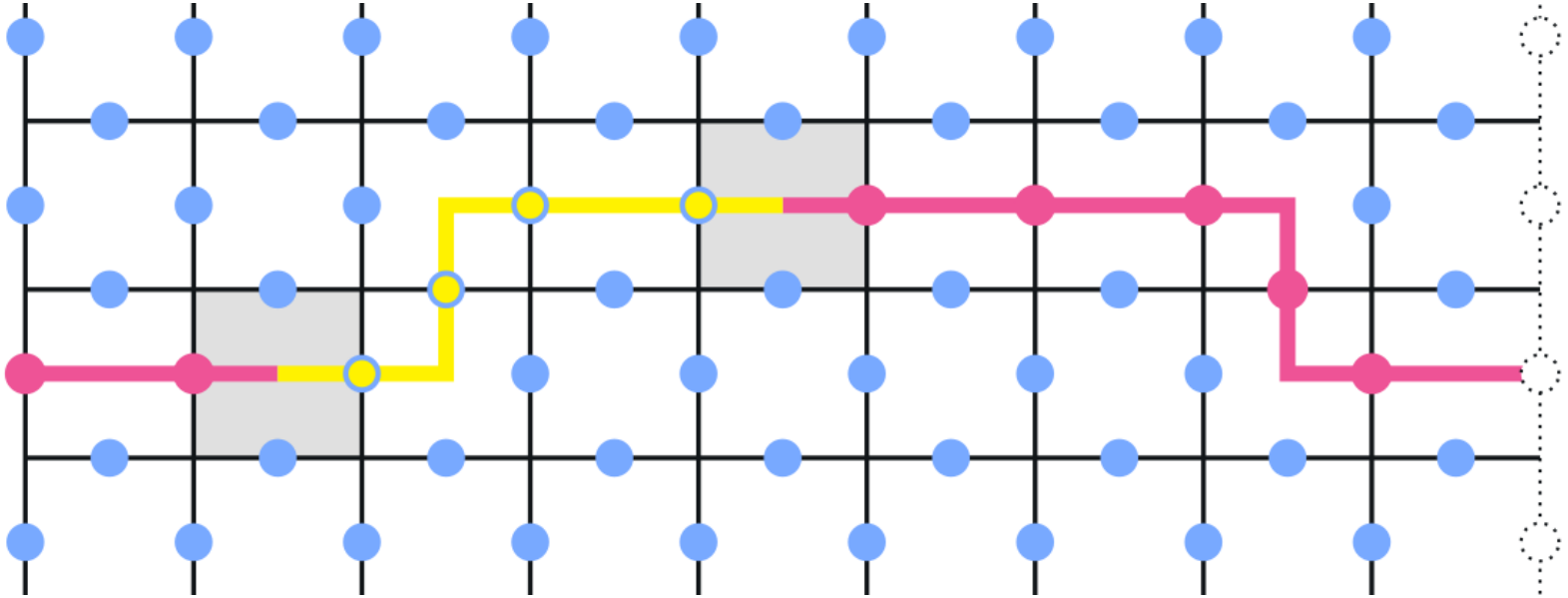
Two paths share the same pair of defects.

E **Actual errors**

Create two endpoint defects.

C **Chosen recovery**

Joins the same two defects.



Left and right boundaries are joined.

Gray: defects before recovery

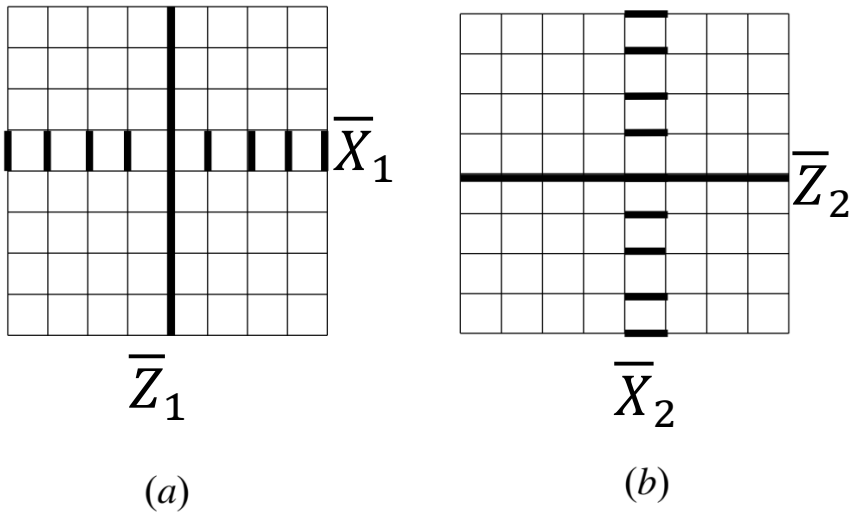
$CE = \overline{X}$

All checks pass.
A logical error remains.

Figure adapted from IBM Quantum Learning, Toric code

Two logical qubits on a torus

The torus has two independent winding directions



(a) (b)

$k = 2, \quad \text{dimension} = 4$

Each dual winding direction supports a logical X loop.

A conjugate Z loop on the original lattice crosses it once.

One crossing gives the logical $XZ = -ZX$ relation.

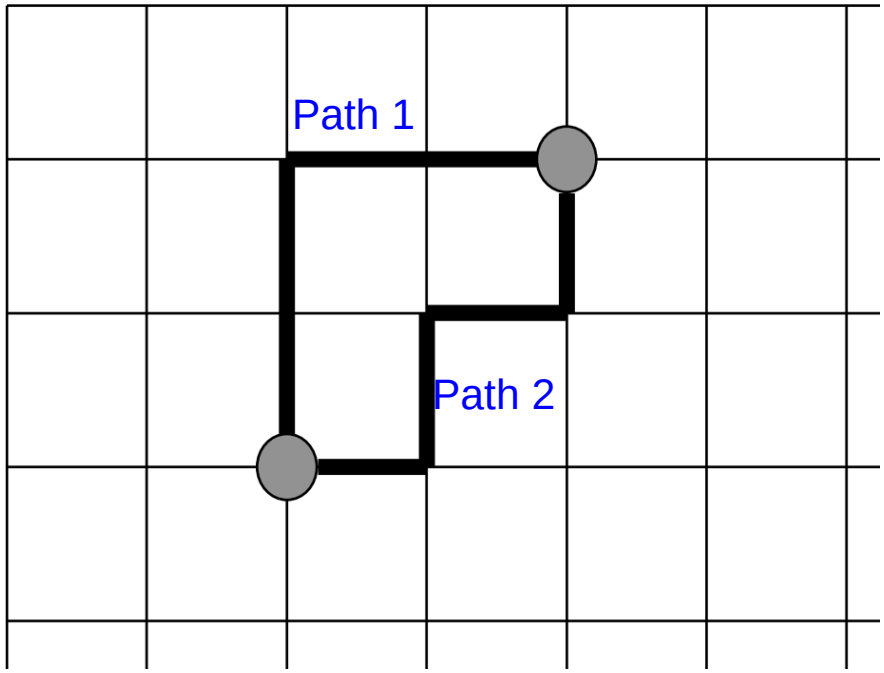
Four basic logical operators form two conjugate pairs, encoding two qubits.

Dennis et al. (2002), Fig. 3

Phase errors use the original lattice

Phase-error chains follow the original lattice edges.

Local toric-code lattice



Data qubits lie on edges.

Two possible error paths

Each bold path is one possible chain of phase flips.

$$E_Z = Z_1 Z_2 \cdots Z_m$$

Defects at vertices

Gray dots mark the two failed X-star checks.

At an interior vertex, two sign changes cancel.

Endpoint star-check outcomes

$$-1, -1$$

Figure: Dennis et al., Topological quantum memory, Fig. 4

Code distance sets the protection scale

The shortest nontrivial logical loop has L edges

$$[[n, k, d]] = [[2L^2, 2, L]]$$

n counts physical data qubits. k counts logical qubits.

correctable-weight guarantee $t = \left\lfloor \frac{d-1}{2} \right\rfloor$

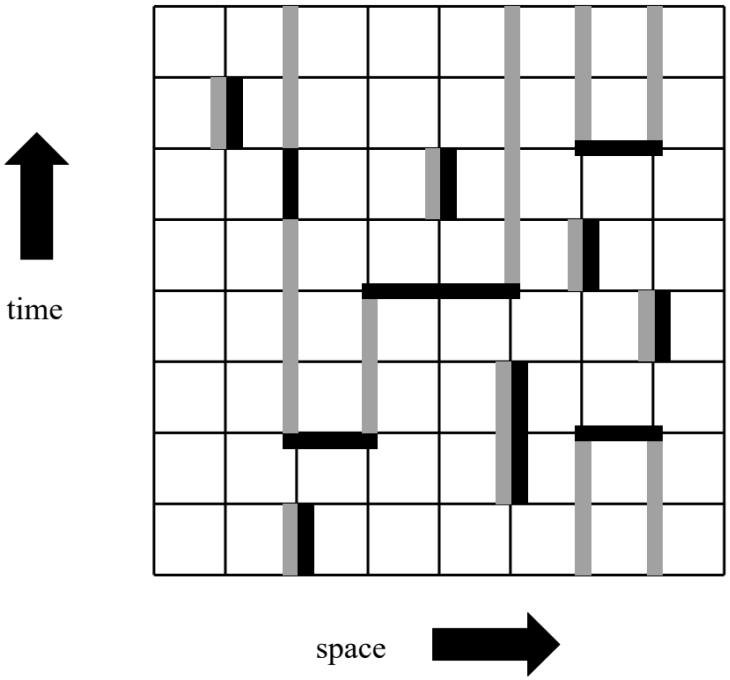
d is the minimum weight of a nontrivial logical operator.

With ideal checks, all errors on at most t qubits are correctable.

$L = 5$: 50 data qubits, 2 logical qubits, distance 5, $t = 2$.

Noisy checks add a time dimension

Real devices repeat syndrome extraction



Data errors can change checks at nearby positions.

Readout errors can create changes across adjacent rounds.

The decoder uses a history of detection events.

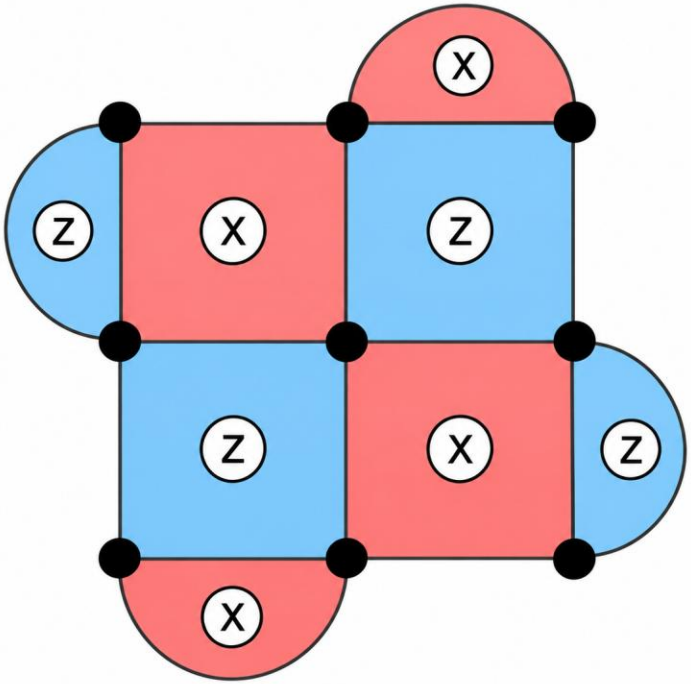
Protection requires good measurements, a suitable circuit and a capable decoder.

Repetition-code illustration. Toric decoding adds a second spatial dimension.

The surface code

Repeated local checks protect one logical qubit.

Rotated distance-3 patch



Black dots: 9 data qubits

White circles: 8 check ancillas

Bulk parity checks

An ancilla checks four neighboring data qubits.

$$S_X = X \otimes X \otimes X \otimes X$$

Detects phase-flip errors

$$S_Z = Z \otimes Z \otimes Z \otimes Z$$

Detects bit-flip errors

Boundary checks act on two data qubits.

Logical errors

A chain across the patch can cause a logical error.

Distance 3 corrects any one data-qubit error with ideal checks.

Toric code and planar surface code

Related codes use different boundary conditions

Feature	Standard toric code	Planar surface-code patch
Geometry	Periodic in both directions	Physical boundaries
Logical information	Two qubits on a torus	Typically one qubit per patch
Core idea	Local X/Z checks	Local X/Z checks
Hardware issue	Periodic connectivity	Boundary checks and layout

A planar surface-code experiment is not automatically a toric-code experiment.

Experiment: error suppression on Willow

Google Quantum AI, Nature 2025

2.14×

error suppression per +2 in distance

Planar surface-code memories at distances 3, 5 and 7.

Logical error fell by 2.14× for each increase of distance by 2.

Distance 7 used 101 physical qubits.

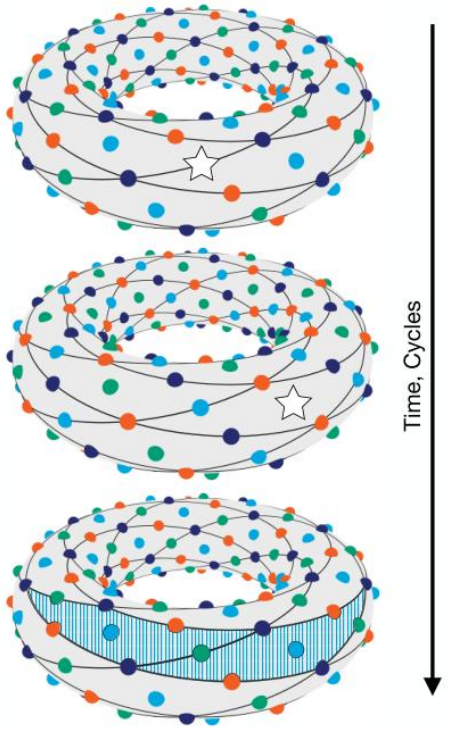
0.143% logical error per cycle

The result demonstrates protected memory and below-threshold scaling.

Google Quantum AI and Collaborators, Nature 638, 920–926 (2025)

Experiment: a toric-code memory with atoms

Atom Computing and Collaborators, June 2026 preprint



Moving neutral atoms implement twisted toric-code connectivity.

Repeated checks include atom replacement and reservoir reloading.

The smaller code reaches 90 cycles of syndrome extraction.

A larger code reduced logical errors in separate comparisons through 8 cycles.

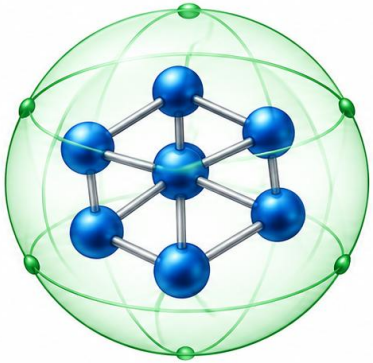
Adapted from arXiv:2606.04079 (2026), Fig. 1, CC BY 4.0. Preprint

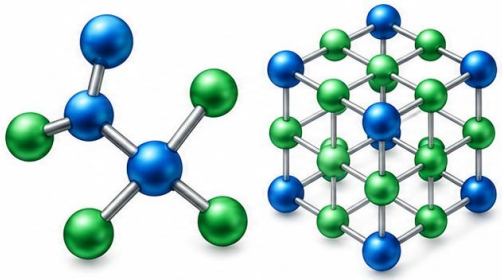
The main ideas

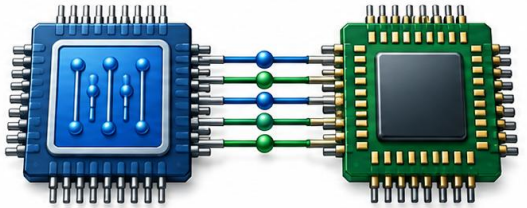
1. Quantum states carry amplitudes, and gates control their interference.
2. Deutsch–Jozsa and Shor extract global structure with carefully chosen readout.
3. Quantum error correction learns errors without learning the logical state.
4. Toric code protects global information using local checks and topology.

Reliable quantum computing combines quantum hardware with classical algorithms.

Outlook for quantum computation







Reliable computation

Protect logical qubits.
Run reliable logical gates.
Lower correction costs.

Useful applications

Chemistry, materials,
and quantum simulation.
Benchmark useful tasks.

Quantum + classical

Classical decoding,
compilers and control
support quantum circuits.

The goal: a useful result with a clear advantage.

Compare total cost and accuracy with the best classical methods.

Appendix: projective measurement

The general form of the measurement postulate

$$P(m) = \langle\psi|P_m|\psi\rangle$$

$$|\psi_m\rangle = \frac{P_m|\psi\rangle}{\sqrt{P(m)}}$$

P_m projects onto the subspace for outcome m.

The projectors are orthogonal and sum to I.

A parity check keeps coherence within its outcome subspace.

For computational-basis measurement, $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$.

Optional

Appendix: the Deutsch–Jozsa amplitude

After the oracle and final Hadamards

$$a(0^n) = 2^{-n} \sum_x (-1)^{f(x)}$$
$$P(0^n) = |a(0^n)|^2$$

Constant: all signs agree, so $a(0^n) = \pm 1$.

Balanced: equal positive and negative terms give $a(0^n) = 0$.

For other outcomes, signs depend on both x and the output string.

The promise converts one interference test into an exact decision.

Optional

Appendix: counting the logical qubits

The two global check relations remove two independent constraints

$$\prod_v A_v = I \qquad \prod_p B_p = I$$

$$k = 2L^2 - (2L^2 - 2) = 2$$

L^2 stars and L^2 plaquettes give $2L^2$ listed checks.

Each edge appears twice in the product of all stars or plaquettes.

Only $2L^2 - 2$ checks are independent on a connected torus.

The code space has dimension $2^k = 4$.

Optional

Appendix: further reading

Foundations and circuits

IBM Quantum Learning, Basics of quantum information

https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information

Deutsch–Jozsa

Deutsch & Jozsa (1992), Cleve et al. (1998)

https://arxiv.org/abs/quant-ph/9708016

Shor's algorithm

P. W. Shor, SIAM J. Comput. 26 (1997)

https://arxiv.org/abs/quant-ph/9508027

Stabilizer error correction

D. Gottesman, PhD thesis (1997)

https://arxiv.org/abs/quant-ph/9705052

Optional

Appendix: toric code and experiments

Toric-code construction

A. Yu. Kitaev, Ann. Phys. 303 (2003)

https://arxiv.org/abs/quant-ph/9707021

Decoding and noisy checks

E. Dennis et al., J. Math. Phys. 43 (2002)

https://arxiv.org/abs/quant-ph/0110143

Below-threshold memory

Google Quantum AI and Collaborators, Nature 638 (2025)

https://www.nature.com/articles/s41586-024-08449-y

Toric-code experiment

Atom Computing and Collaborators (2026), preprint

https://arxiv.org/abs/2606.04079

Optional